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$$a_{(p+q)} = \frac{p+q}{pq}$$

In H.P:

$$a_{(p+q)} = \frac{pq}{p+q} \quad \text{proved!}$$

Ex# 4.9 Your Success is our win

(i) Insert H.M b/w:

(i) $\frac{1}{64}$ and $\frac{1}{81}$ Sol

$$\therefore H = \frac{2ab}{a+b}$$

$$\text{here, } a = \frac{1}{64} \text{ \& } b = \frac{1}{81}$$

$$\Rightarrow H = \frac{2(\frac{1}{64})(\frac{1}{81})}{\frac{1}{64} + \frac{1}{81}}$$

$$\Rightarrow H = \frac{2(\frac{1}{64})(\frac{1}{81})}{\frac{81+64}{(81)(64)}}$$

$$\Rightarrow H = \frac{2}{145}$$

Alternative Method:

In A.P:

64 and 81

$$A = \frac{a+b}{2}$$

$$A = \frac{64+81}{2}$$

$$A = 145/2$$

In H.P:

 $\frac{2}{145}$ is req. H.M.(ii) $\frac{1}{5}$ and $\frac{1}{17}$ Sol

In A.P: 5 and 17

$$\therefore A = \frac{a+b}{2}$$

$$\Rightarrow A = \frac{5+17}{2}$$

$$\Rightarrow A = \frac{22}{2} = 11$$

In H.P

 $\frac{1}{11}$ is the req. H.M.

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(iii) $3+\sqrt{2}$ and $3-\sqrt{2}$ Sol

$$\therefore H = \frac{2ab}{a+b}$$

Here, $a = 3+\sqrt{2}$ & $b = 3-\sqrt{2}$ (iv) $1/7$ and $1/11$

$$\therefore H = \frac{2(3+\sqrt{2})(3-\sqrt{2})}{3+\sqrt{2}+3-\sqrt{2}}$$

$$3+\sqrt{2}+3-\sqrt{2}$$

$$\Rightarrow H = \frac{2[(3)^2 - (\sqrt{2})^2]}{6}$$

$$\Rightarrow H = \frac{9-2}{3} = \frac{7}{3}$$

It is the required H.M. In H.P:

$$\Rightarrow H = 2(4+9)/4$$

$$\Rightarrow H = 26/4 = 13/2$$

It is the req H.M.

Sol

In A.P: 7 and 11.

$$\therefore A = \frac{a+b}{2}$$

$$\Rightarrow A = \frac{7+11}{2} = \frac{18}{2} = 9$$

 $1/9$ is the required H.M.(iv) $2+3i$ and $2-3i$ Sol

$$\therefore H = \frac{2ab}{a+b}$$

here, $a = 2+3i$ and $b = 2-3i$

$$\therefore H = \frac{2(2+3i)(2-3i)}{2+3i+2-3i}$$

$$2+3i+2-3i$$

$$\Rightarrow H = \frac{2[(2)^2 - (3i)^2]}{4}$$

$$2+3i+2-3i$$

$$\Rightarrow H = \frac{2(4-9i^2)}{4}$$

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(vi) 3 and 7

Sol

$$\text{In A.P. } \therefore H = \frac{2ab}{a+b}$$

here, $a = 3$ and $b = 7$

$$\therefore H = \frac{2(3)(7)}{3+7}$$

$$3+7$$

$$\Rightarrow H = \frac{2(21)}{10} = \frac{21}{5}$$

It is the required H.M.

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(2) Insert four H.Ms b/w

4 and 20

Sollet H_1, H_2, H_3 and H_4 be

the 4 H.Ms.

In A.P:

$$\therefore H_1 = \frac{1}{4} \text{ and } \frac{1}{20}$$

$$\therefore d = \frac{b-a}{n+1}$$

$$\Rightarrow d = \frac{\frac{1}{20} - \frac{1}{4}}{5+1}$$

$$\Rightarrow d = -\frac{1}{25}$$

Now,

$$A_1 = a + d = \frac{1}{4} - \frac{1}{25} = \frac{21}{100}$$

$$A_2 = \frac{1}{4} + 2(-\frac{1}{25}) = \frac{17}{100}$$

$$A_3 = \frac{1}{4} + 3(-\frac{1}{25}) = \frac{13}{100}$$

$$A_4 = a + 4d = \frac{1}{4} + 4(-\frac{1}{25}) = \frac{9}{100}$$

In H.P:

$$\frac{100}{21}, \frac{100}{17}, \frac{100}{13}, \frac{100}{9}$$

are the required 4 H.Ms.

(ii) $\frac{1}{9}$ and $\frac{1}{81}$ Sol

In A.P:

9 and 81

$$\therefore d = \frac{b-a}{n+1}$$

here, $a = 9$ and $b = 81$

$$\Rightarrow d = \frac{81-9}{5+1}$$

$$\Rightarrow d = \frac{72}{5}$$

Now,

$$A_1 = a + d = 9 + \frac{72}{5} = \frac{117}{5}$$

$$A_2 = a + 2d = 9 + 2(\frac{72}{5}) = \frac{189}{5}$$

$$A_3 = a + 3d = 9 + 3(\frac{72}{5}) = \frac{261}{5}$$

$$A_4 = a + 4d = 9 + 4(\frac{72}{5}) = \frac{333}{5}$$

In H.P:

$$\frac{5}{117}, \frac{5}{189}, \frac{5}{261}, \frac{5}{333}$$

are the req 4 H.Ms.

(3) Insert five H.Ms. b/w

 $\frac{1}{4}$ and $\frac{1}{24}$ Sol

In A.P:

4 and 24.

$$\therefore d = \frac{b-a}{n+1}$$

here,

$$a = 4, b = 24 \text{ \& } n = 5$$

$$\therefore d = \frac{24-4}{5+1}$$

$$\Rightarrow d = \frac{20}{6} = \frac{10}{3}$$

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Now,

$$A_1 = a + d = 4 + 10/3 = 22/3$$

$$A_2 = a + 2d = 4 + 2(10/3) = 32/3$$

$$A_3 = a + 3d = 4 + 3(10/3) = 14$$

$$A_4 = a + 4d = 4 + 4(10/3) = 52/3$$

$$A_5 = a + 5d = 4 + 5(10/3) = 62/3$$

In H.P:

$$3/22, 3/32, 1/14, 3/52, 3/62$$

are the req 5 H.Ms.

- (4) Prove the square of the G.M of 2 numbers equal to the product of the A.M and H.M of 2 numbers.

Sol

let a and b be 2 Nos.

For A.M:

$$\therefore A = a + b/2$$

For G.M:

$$\therefore G = \pm \sqrt{ab}$$

For H.M.

$$\therefore H = 2ab/a+b$$

A/c to given condition.

$$\Rightarrow G^2 = (A)(H)$$

$$\Rightarrow (\pm \sqrt{ab})^2 = (a+b/2) \left(\frac{2ab}{a+b} \right)$$

$$\Rightarrow ab = 2ab/2$$

$$\Rightarrow ab = ab$$

proved!

- (5) Find b if 5 is H.M b/w 2 and b.

Sol

$$\therefore H = 2ab/a+b$$

here,

$$a = 2, H = 5 \text{ and } b = b.$$

Now,

$$\Rightarrow 5 = \frac{2(2)b}{2+b}$$

$$\Rightarrow 10 + 5b = 4b$$

$$\Rightarrow b = -10$$

BQ

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- ⑦ Find A.M, G.M & H.M b/w 1 and 9. Also show that $(A)(H) = G^2$

Sol

For A.M:

$$\therefore A = \frac{a+b}{2}$$

$$\Rightarrow A = \frac{1+9}{2}$$

$$\Rightarrow A = 5$$

For H.M:

$$\therefore H = \frac{2ab}{a+b}$$

$$\Rightarrow H = \frac{2(1)(9)}{1+9}$$

$$\Rightarrow H = \frac{18}{10} = \frac{9}{5}$$

For G.M:

$$\therefore G = \pm \sqrt{ab}$$

$$\Rightarrow G = \pm \sqrt{(9)(1)}$$

$$\Rightarrow G = \pm 3$$

$$\Rightarrow AH = G^2$$

$$\Rightarrow (5)\left(\frac{9}{5}\right) = (\pm 3)^2$$

$$\Rightarrow 9 = 9 \quad \text{proved!}$$

- ⑧ A.M of 2 no.s is 8 & H.M is 6 - Find numbers.

Sol

A/c to given condition:

$$\Rightarrow \text{A.M} = 8 \quad \text{--- ①}$$

$$\Rightarrow \text{H.M} = 6 \quad \text{--- ②}$$

Take ①

$$\Rightarrow \frac{a+b}{2} = 8$$

$$\Rightarrow a+b = 16$$

$$\Rightarrow a = 16 - b \quad \text{--- ③}$$

Take ②

$$\Rightarrow \frac{2ab}{a+b} = 6$$

$$\therefore a+b = 16$$

$$\Rightarrow \frac{2ab}{16} = 6$$

$$\Rightarrow ab = 48$$

Put a in above eq.

$$\Rightarrow b(16-b) = 48$$

$$\Rightarrow 16b - b^2 = 48$$

$$\Rightarrow b^2 - 16b + 48 = 0$$

$$\Rightarrow b^2 - 12b - 4b + 48 = 0$$

$$\Rightarrow b(b-12) - 4(b-12) = 0$$

$$\Rightarrow (b-12)(b-4) = 0$$

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either,

$$b = 12 \text{ or } b = 4$$

Put b in (3)At $b = 4$ we get,

$$\Rightarrow a = 16 - 4 = 12$$

At $b = 12$ we get,

$$\Rightarrow a = 16 - 12 = 4$$

Thus the req 2 numbers are 4 and 12.

Q The H.M of two numbers is $\frac{24}{5}$ and G.M = 6 numbers. Sol

A/c to given Condition:

$$\Rightarrow \text{H.M} = \frac{24}{5} \text{ — (1)}$$

$$\Rightarrow \text{G.M} = 6 \text{ — (2)}$$

Take (1)

let a and b be 2 nos.

$$\Rightarrow \frac{2ab}{a+b} = \frac{24}{5} \text{ — (3)}$$

Take (2):

$$\Rightarrow \sqrt{ab} = 6$$

$$\Rightarrow ab = 36 \text{ — (4)}$$

Put ab in (3)

$$\therefore (3) \Rightarrow \frac{2(36)}{a+b} = \frac{24}{5}$$

$$\Rightarrow \frac{72 \times 5}{24} = a+b$$

$$\Rightarrow a+b = 15$$

$$\Rightarrow a = 15 - b \text{ — (5)}$$

Put a in (4)

$$\therefore (4) \Rightarrow b(15-b) = 36$$

$$\Rightarrow 15b - b^2 = 36$$

$$\Rightarrow b^2 - 15b + 36 = 0$$

$$\Rightarrow b^2 - 12b - 3b + 36$$

$$\Rightarrow b(b-12) - 3(b-12)$$

$$\Rightarrow (b-12)(b-3) = 0$$

Either,

$$b = 12 \text{ or } b = 3$$

Put b in (5)At $b = 12$ we

$$\Rightarrow a = 15 - 12 = 3$$

At $b = 3$ we get,

$$\Rightarrow a = 15 - 3 = 12$$

Thus 3 and 12 are the req 2 numbers

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5) Find n so that $\frac{x^{n-5} + y^{n-5}}{x^{n-6} + y^{n-6}}$ may be H.M b/w x and y .

Sol

$$\therefore H.M = \frac{2xy}{x+y}$$

$$\Rightarrow \frac{2xy}{x+y} = \frac{x^{n-5} + y^{n-5}}{x^{n-6} + y^{n-6}}$$

$$\Rightarrow 2xy(x^{n-6} + y^{n-6}) = (x+y)(x^{n-5} + y^{n-5})$$

$$\Rightarrow 2x^{n-5}y + 2xy^{n-5} = x^{n-4} + xy^{n-5} + yx^{n-5} + y^{n-4}$$

$$\Rightarrow x^{n-5}y + xy^{n-5} = x^{n-4} + y^{n-4}$$

$$\Rightarrow x^{n-5}y - x^{n-4} = y^{n-4} - xy^{n-5}$$

$$\Rightarrow x^{n-5}(y/x) = y^{n-5}(y/x)$$

$$\Rightarrow \frac{x^{n-5}}{y^{n-5}} = 1$$

$$\Rightarrow \left(\frac{x}{y}\right)^{n-5} = \left(\frac{x}{y}\right)^0$$

Thus,

$$\Rightarrow n-5 = 0$$

$$\Rightarrow n = 5$$